



ISMRM & ISMRT
ANNUAL MEETING & EXHIBITION

Honolulu, Hawai'i, USA 10-15 MAY 2025



ISMRM 2025

MR Artifacts Game Show

**Classic Artifacts in Radial MRI and
The Semi-convergence Behavior of CG-SENSE**

Hung Do, PhD MSEE
Canon Medical Systems USA



ISMRM 2025 – Honolulu, HI

ISMIRM 2025

MR Artifacts Game Show



STAY CONNECTED DURING THE MEETING:





Declaration of Financial Interests or Relationships

Speaker Name: **Hung Do, PhD MSEE**

I have the following financial interest or relationship to disclose with regard to the subject matter of this presentation:

Company Name: **Canon Medical Systems USA**

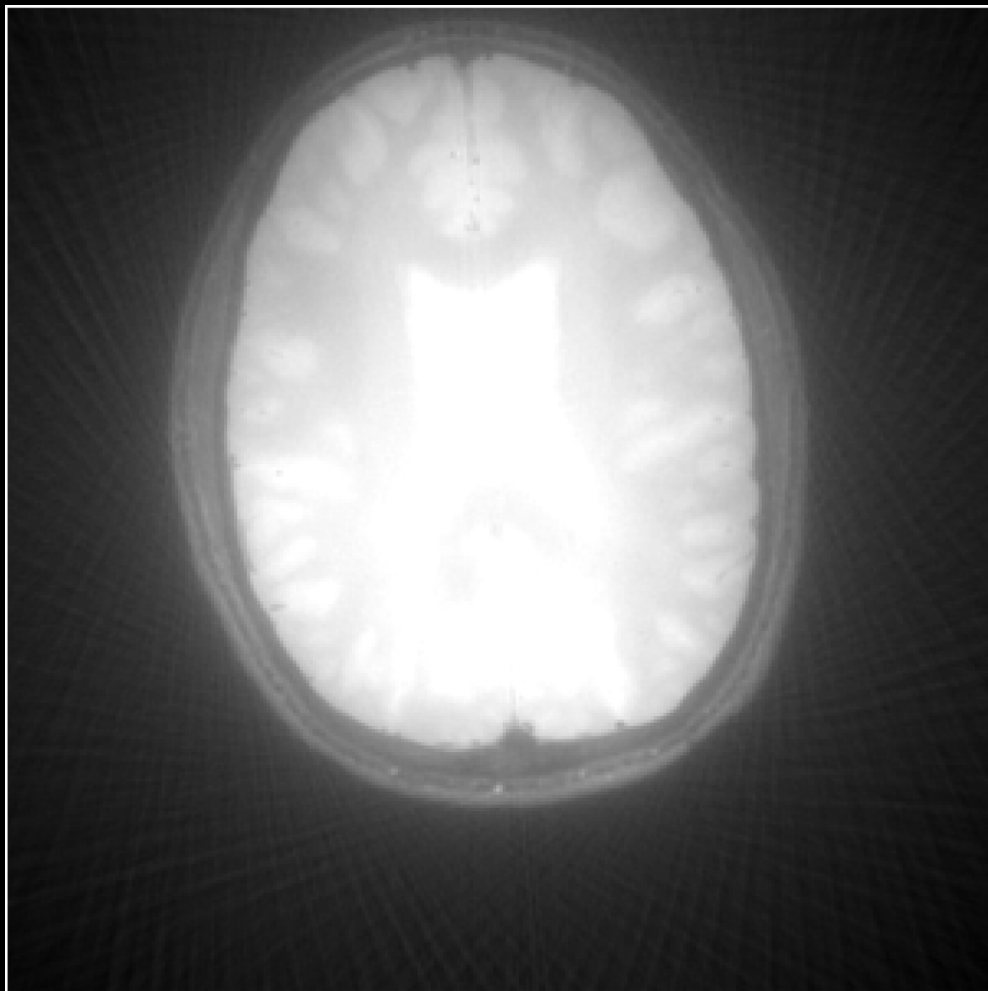
Type of Relationship: **Employee**



The *halo* artifact!



Digi Venkat, Wikipedia.org



Images from: Hung Do, Canon Medical USA

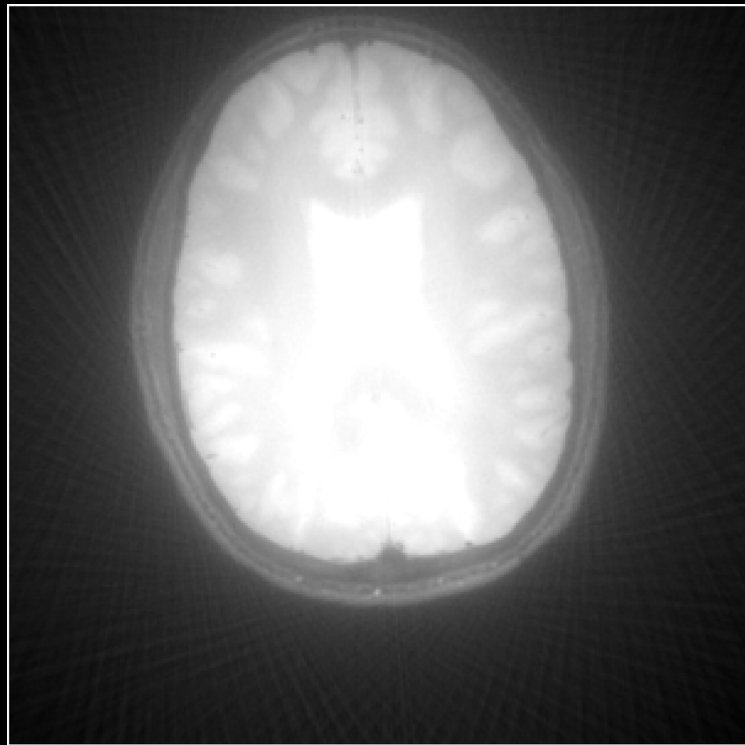


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What causes the *halo* artifact?

- A. Corrupted low-frequency k-space
- B. Corrupted high-frequency k-space
- C. Forget to apply intensity correction
- D. Forget to apply density correction

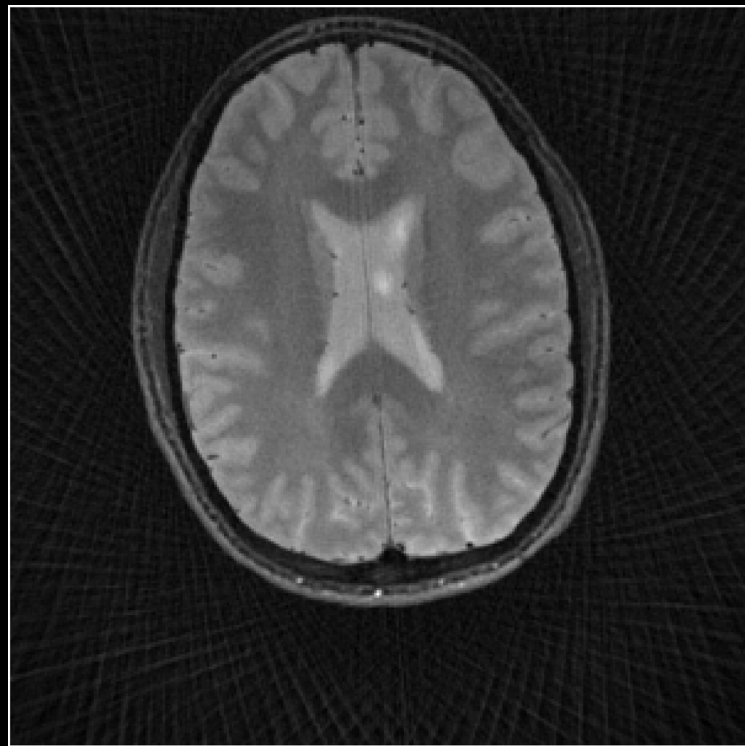
2D single-slice radial imaging with
gridding reconstruction



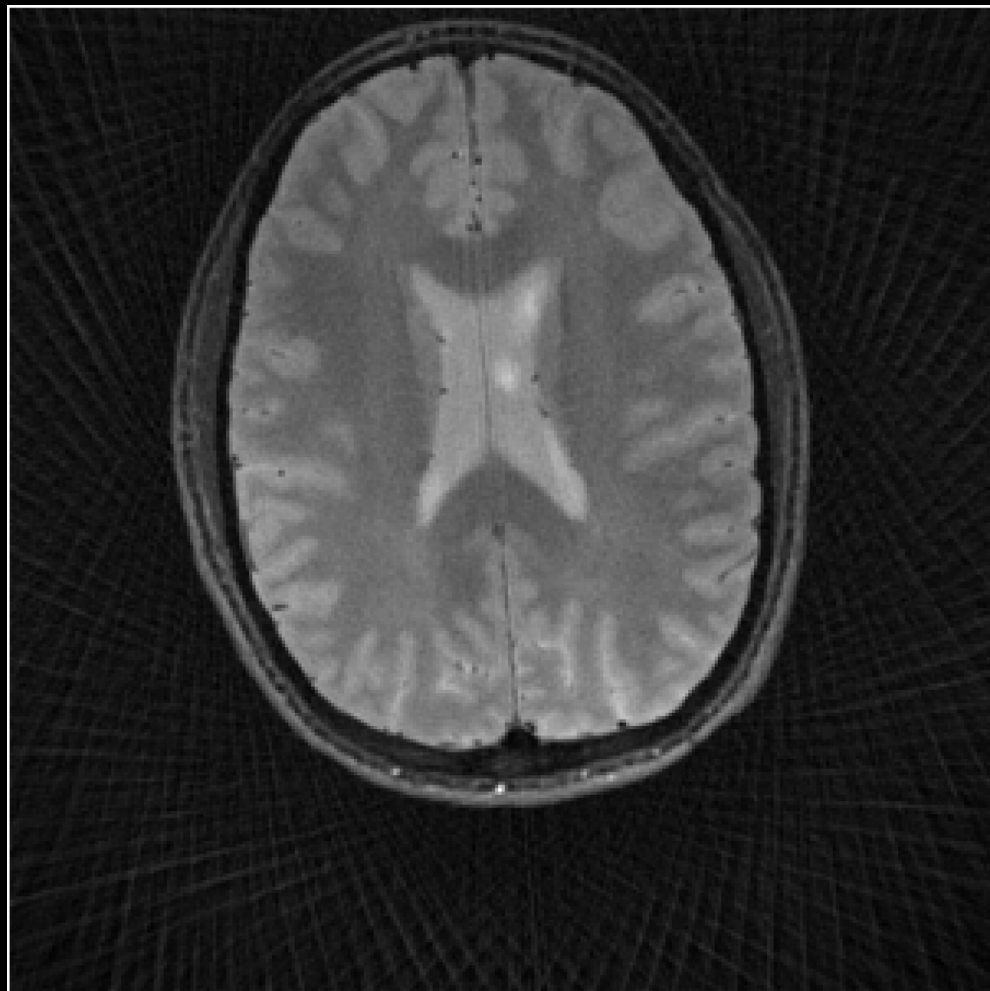
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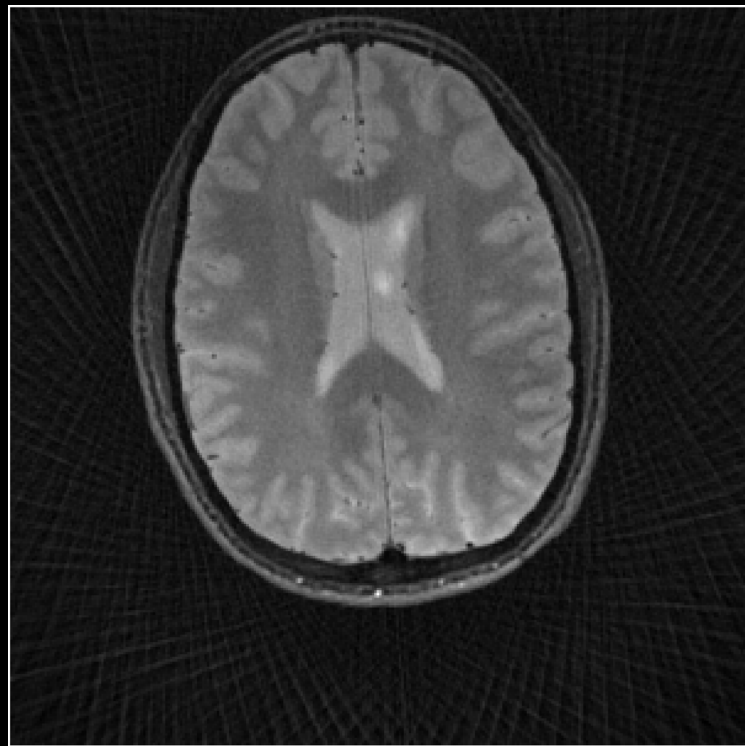
The *streaking* artifact!



What causes the *streaking* artifact?

- A. Head rotating motions
- B. Over-sampled k-space center
- C. Under-sampled k-space data
- D. Gridding kernel is too narrow

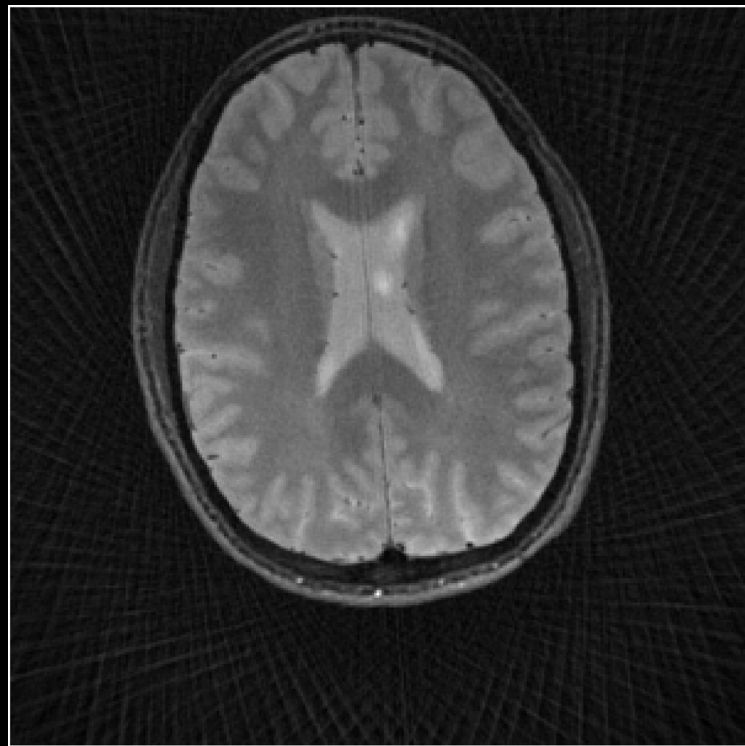
2D single-slice radial imaging with
gridding reconstruction



What causes the *streaking* artifact?

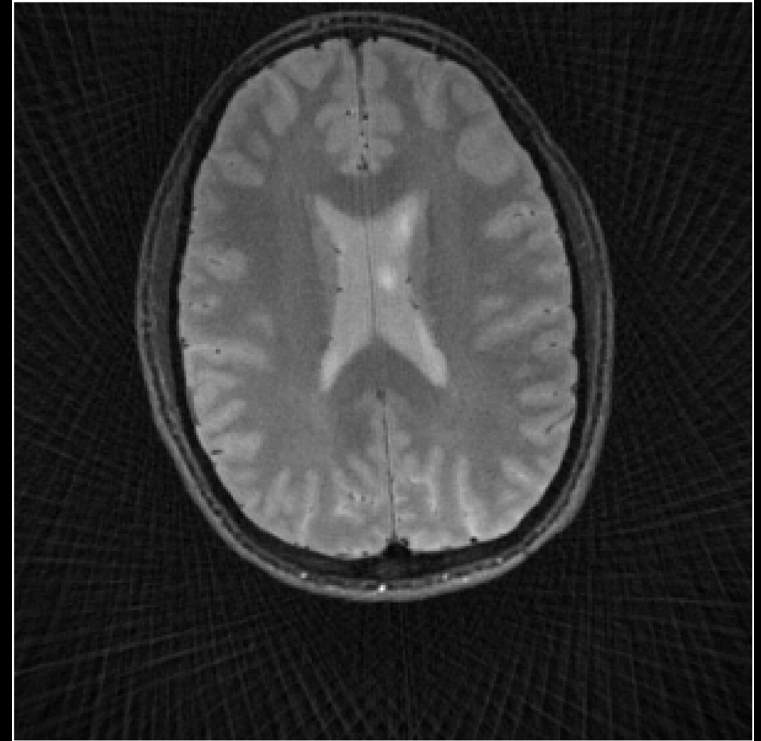
- A. Head rotating motions
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2D single-slice radial imaging with
gridding reconstruction



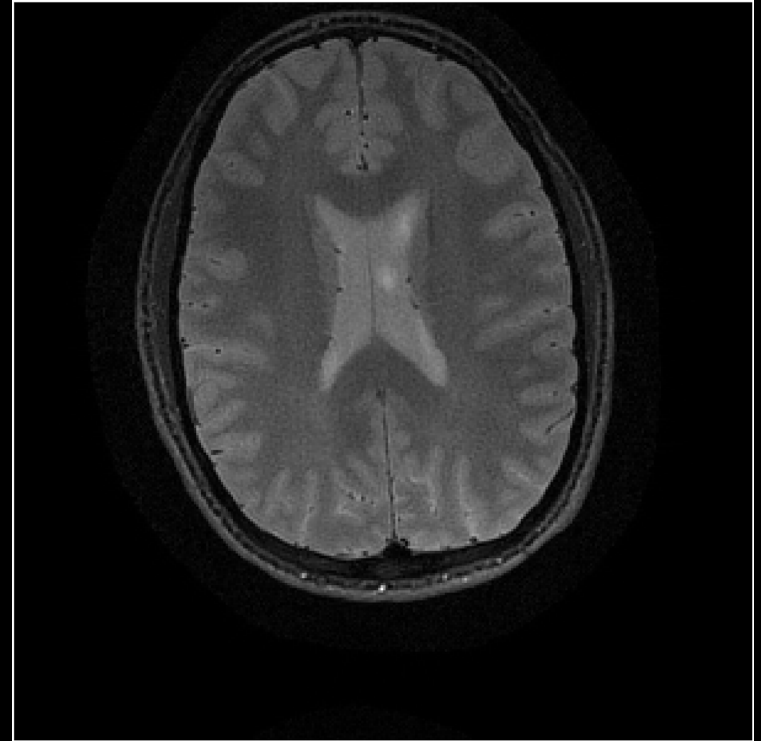
How can the *streaking* artifact be resolved?

- A. Use **SENSE** (Sensitivity encoding)
- B. Use **CG-SENSE** (Conjugate Gradient SENSE)
- C. Use **GRAPPA** (Generalized Auto calibrating Partial Parallel Acquisition)
- D. Use **NuFFT** (Non-uniform Fast Fourier Transform)



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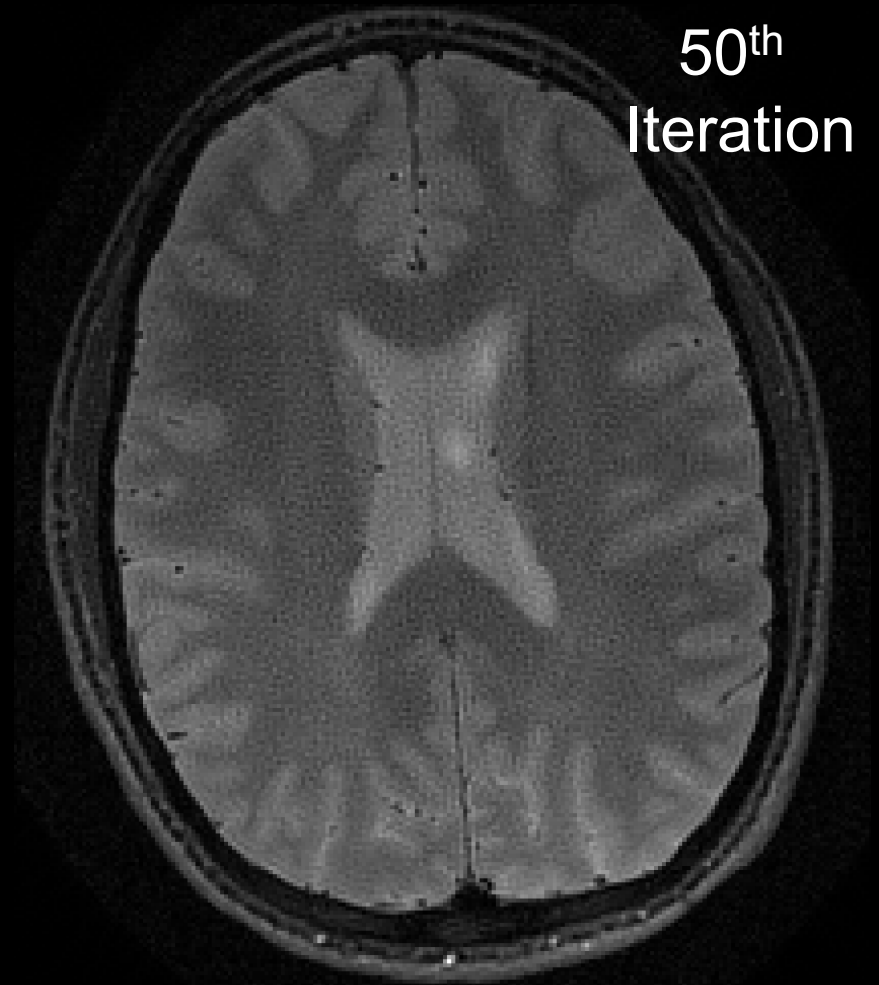
Bonus: Challenging Artifacts



What is the Artifact?

- A. Remnants of streaking artifacts
- B. Over-suppressed background noise
- C. Noise amplification
- D. Under smoothing

50th
Iteration



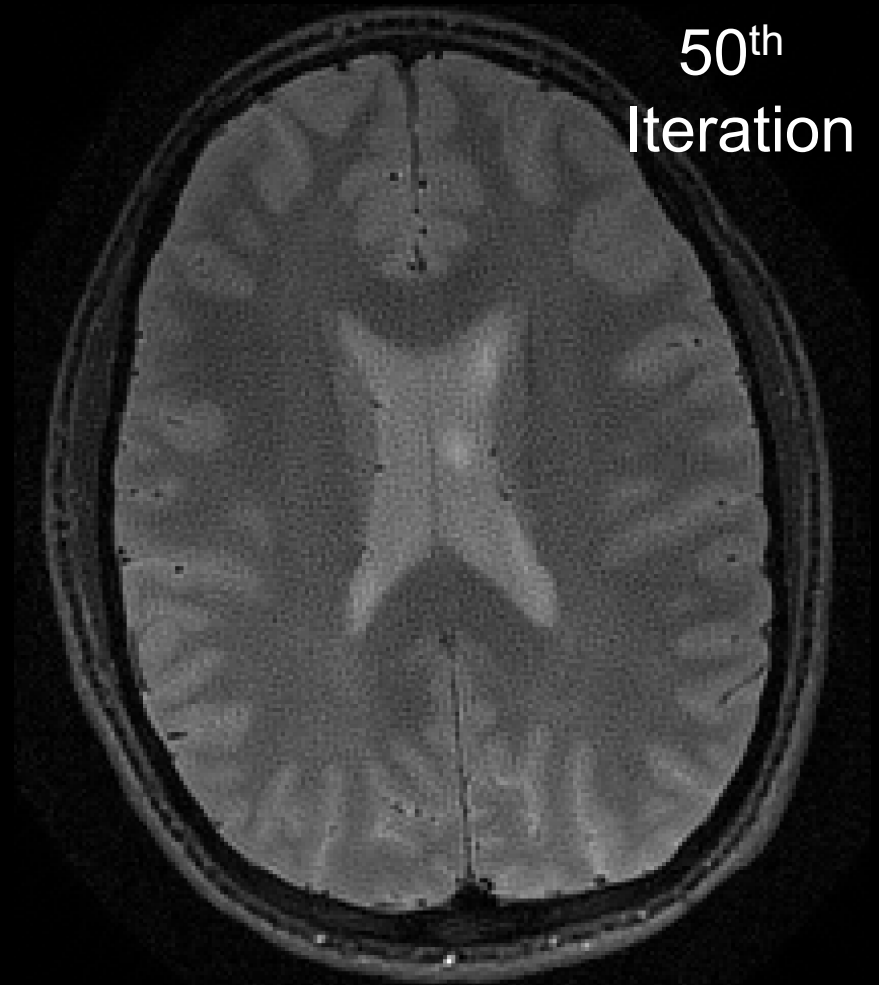
CG-SENSE Recon in action!



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50th
Iteration



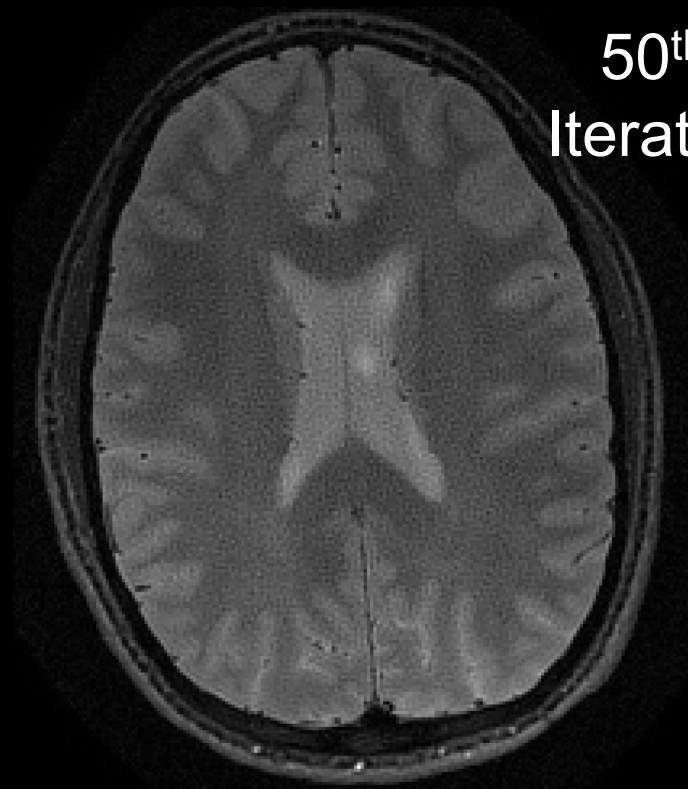
CG-SENSE Recon in action!



How can the artifact be resolved?

- A. Decrease the number of iterations
- B. Increase the number of iterations
- C. Remove regularization
- D. Add noise pre-whitening step to the reconstruction pipeline

50th
Iteration



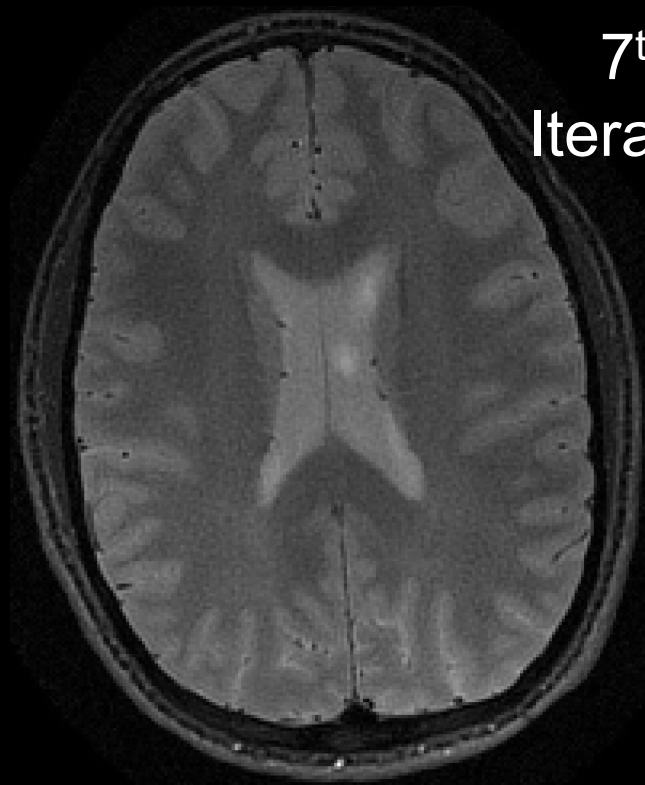
CG-SENSE Recon in action!



How can the artifact be resolved?

- A. Decrease the number of iterations**
- B. Increase the number of iterations
- C. Remove regularization
- D. Add noise pre-whitening step to the reconstruction pipeline

7th
Iteration



CG-SENSE Recon in action!



Artifacts Explanation

Hung Do, PhD MSEE
Canon Medical Systems USA





Declaration of Financial Interests or Relationships

Speaker Name: **Hung Do, PhD MSEE**

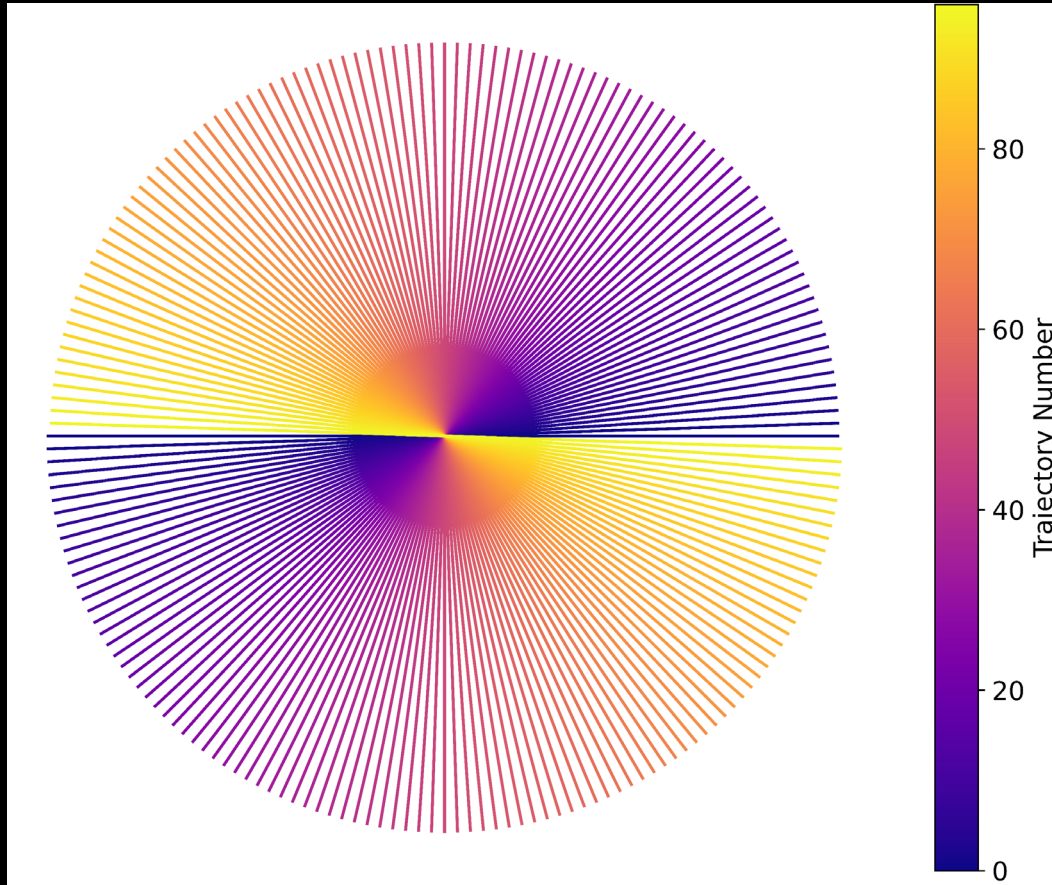
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Type of Relationship: **Employee**



Radial k-space trajectory



Non-uniform k-space data:

- Denser low-frequency samples (center)
- Sparser high-frequency samples (periphery)

The non-uniformity must be compensated before reconstruction to avoid the *halo* (low-frequency) artifact.



Gridding reconstruction

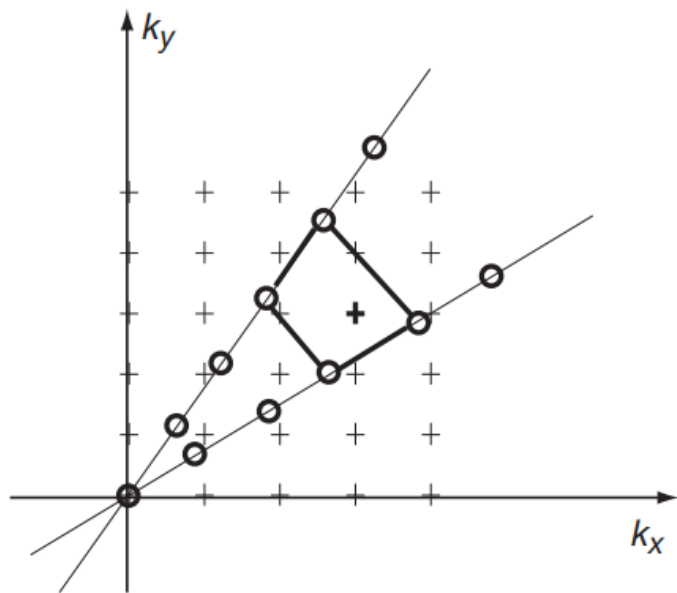


Figure 5.5: Grid-driven interpolation for a projection data set. Data samples lie on diameters in k -space. In this example the surrounding four data samples (o's) are located for each grid point (+'s), and a value at the grid point determined by bilinear interpolation.

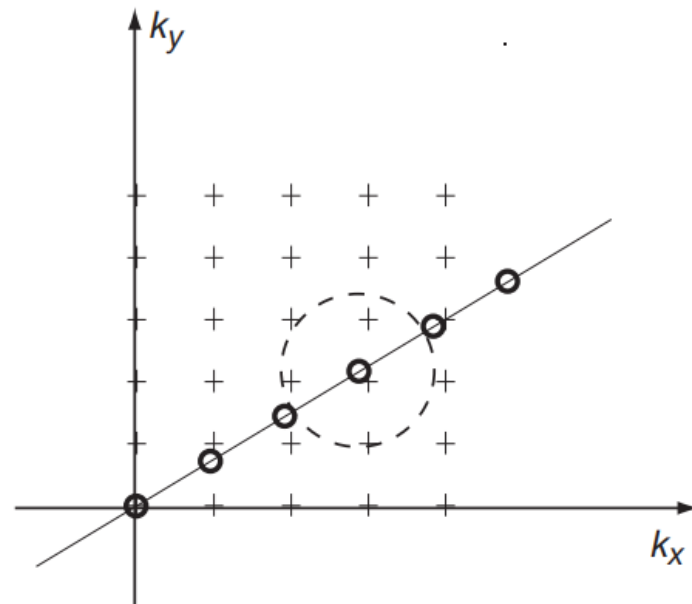


Figure 5.6: Data-driven interpolation for a projection data set. Again, data samples lie on diameters in k -space. Each data point is conceptually considered to be convolved with a small kernel, and the value of that convolution added to the adjacent k -space grid points.



Gridding reconstruction

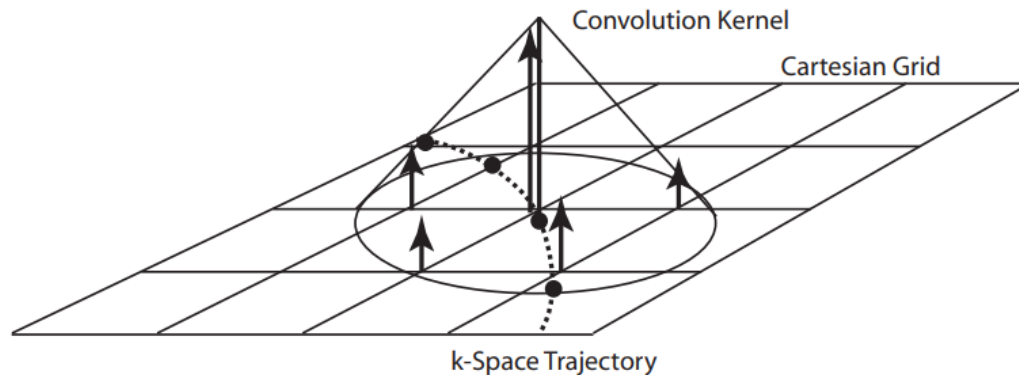
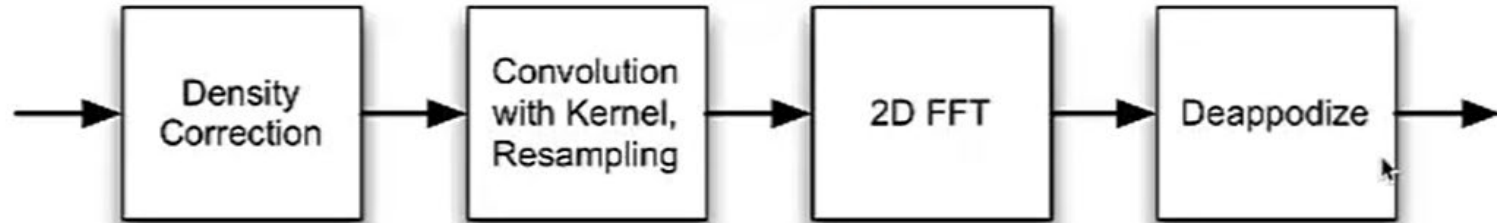


Figure 5.7: Basic gridding idea. The data samples line on some trajectory through k-space (dashed line). Each data point is conceptually convolved with a gridding kernel, and that convolution evaluated at the adjacent grid points.

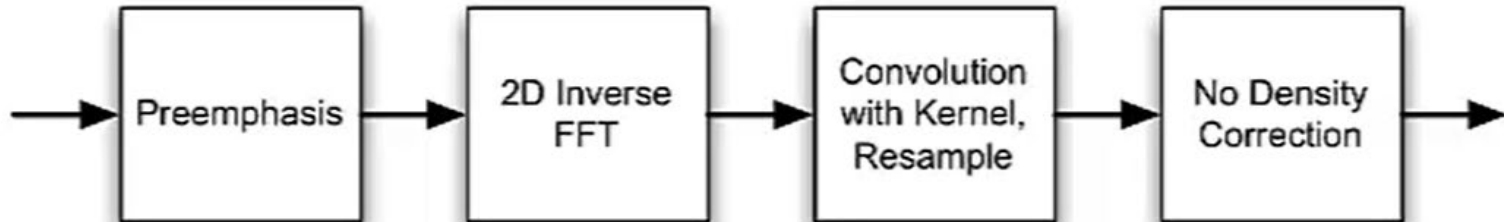


Gridding & Inverse Gridding

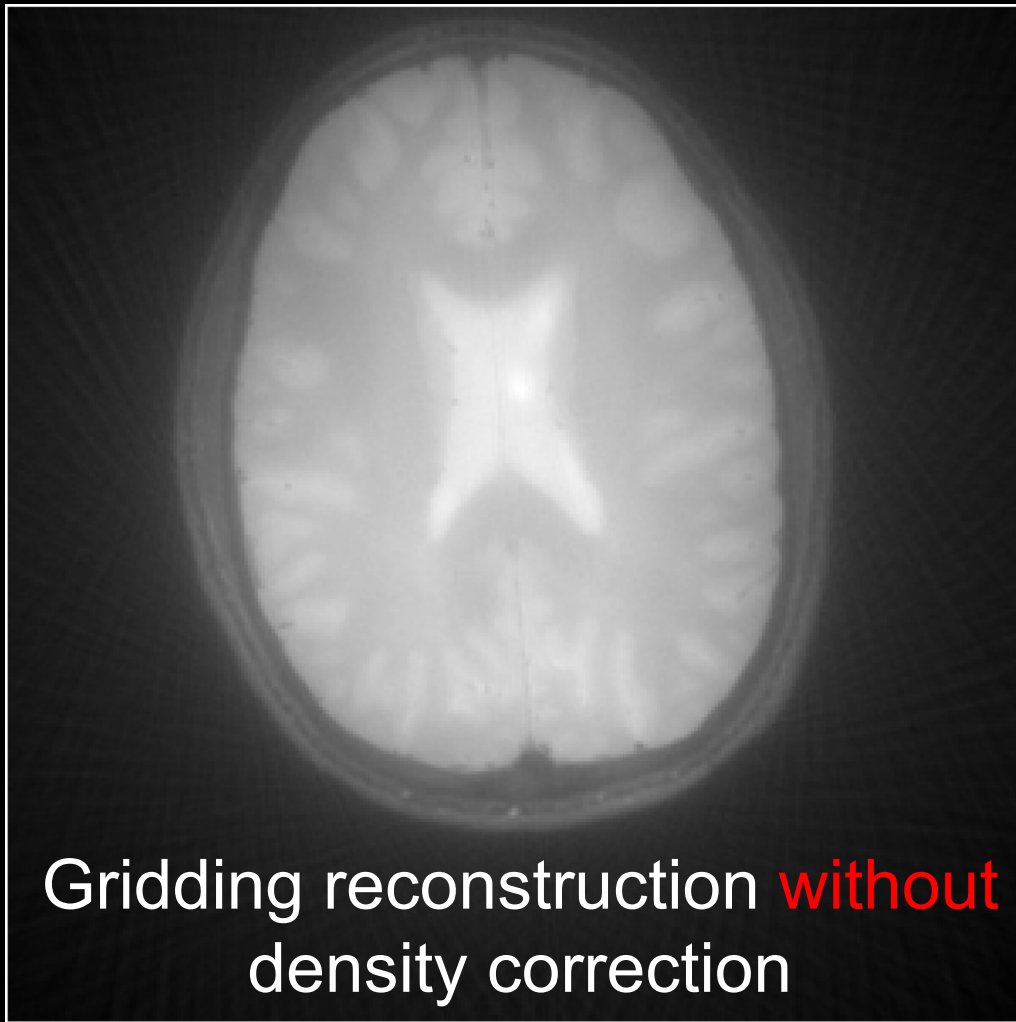
Gridding



Inverse Gridding



The *halo*
(low-frequency)
artifact



Gridding reconstruction **without**
density correction

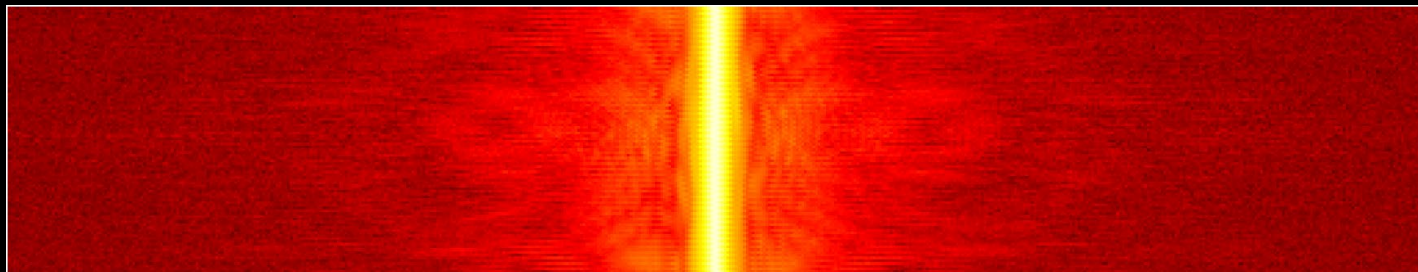


Radial
spoke #

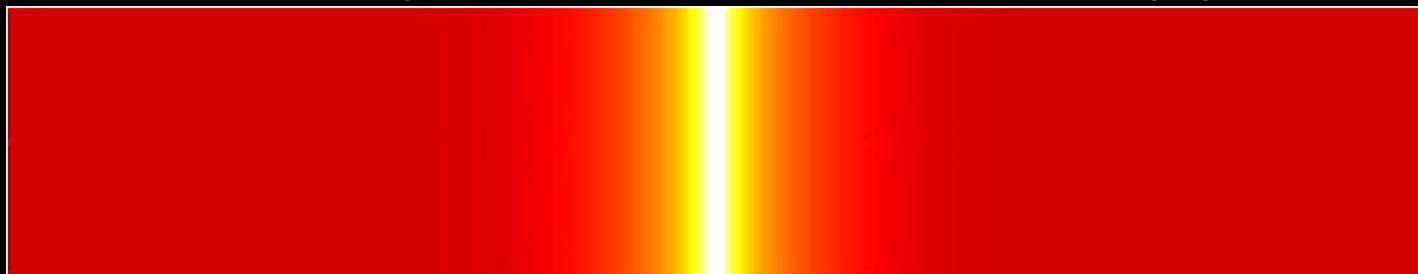


Readout

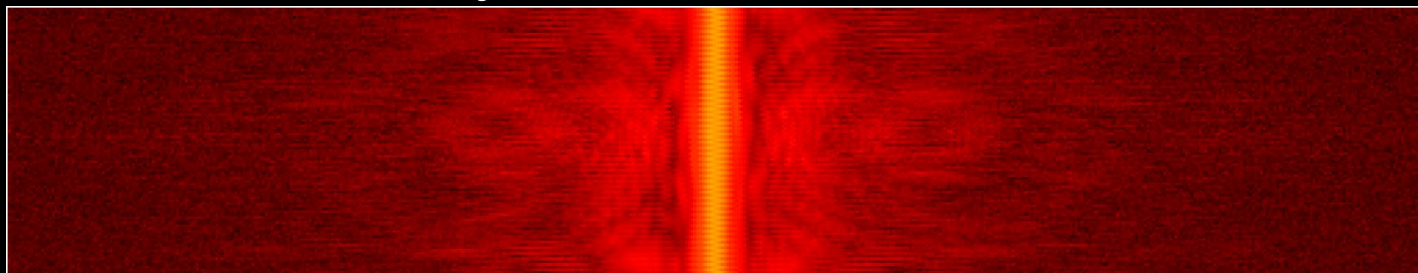
Raw k-space data (**m**)



Density compensation function (**d**)



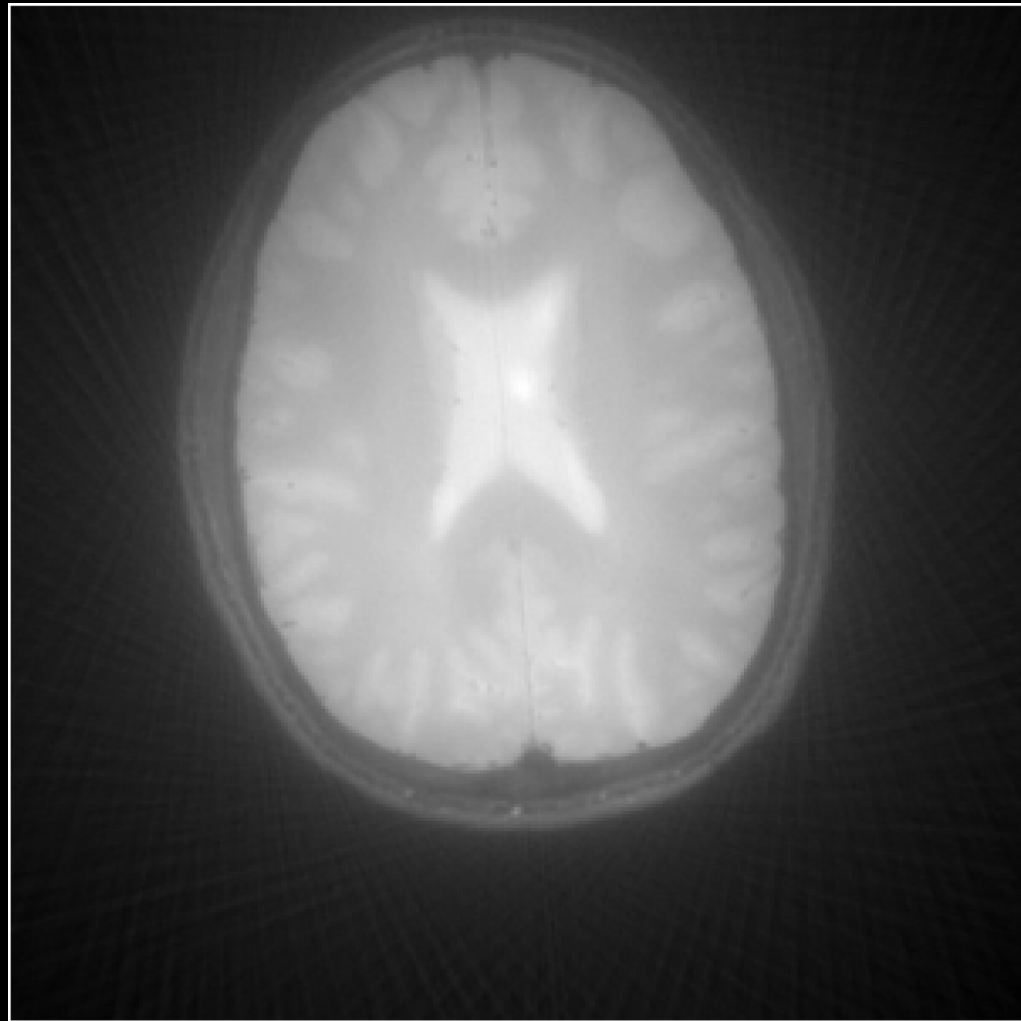
Density corrected data = $\mathbf{m/d}$



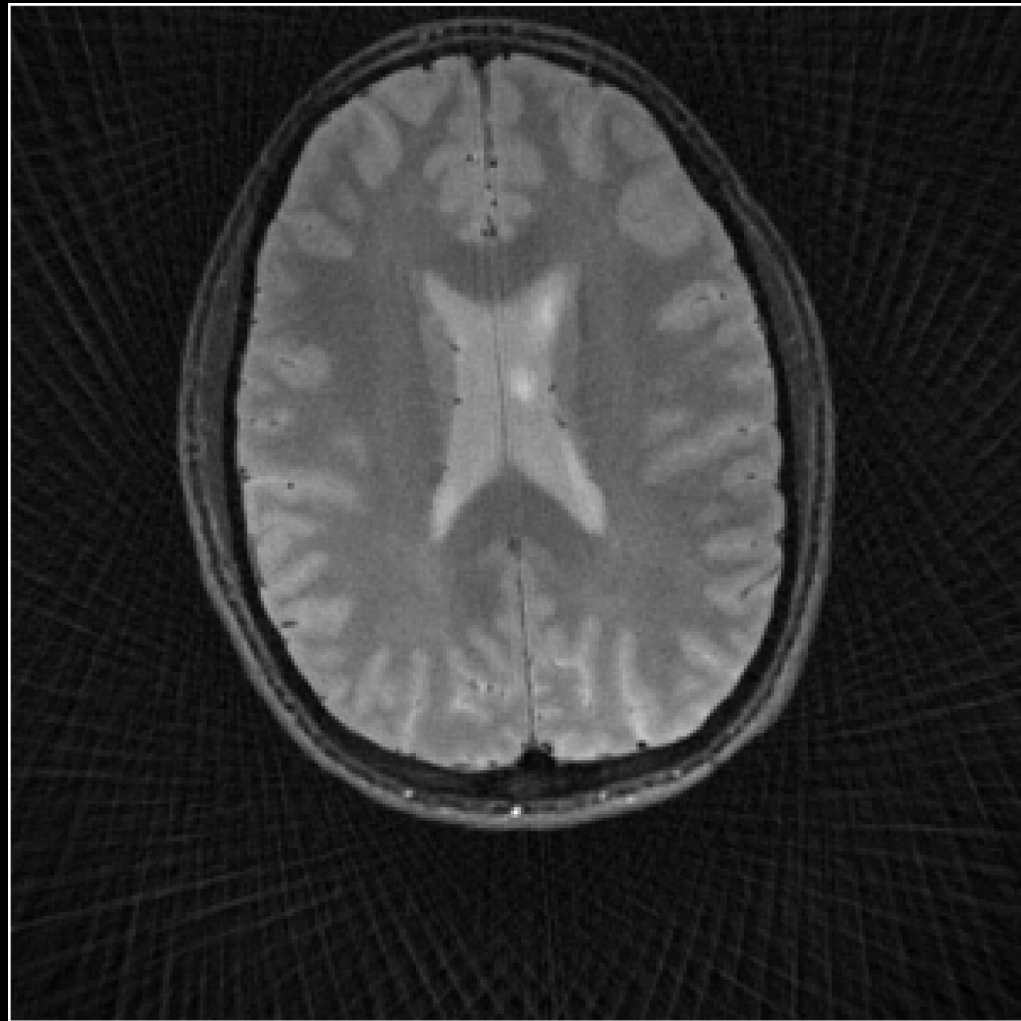
Density
correction



Gridding reconstruction
without
density correction
(halo artifact)



Gridding reconstruction
with
density correction
(streaking and blurring)

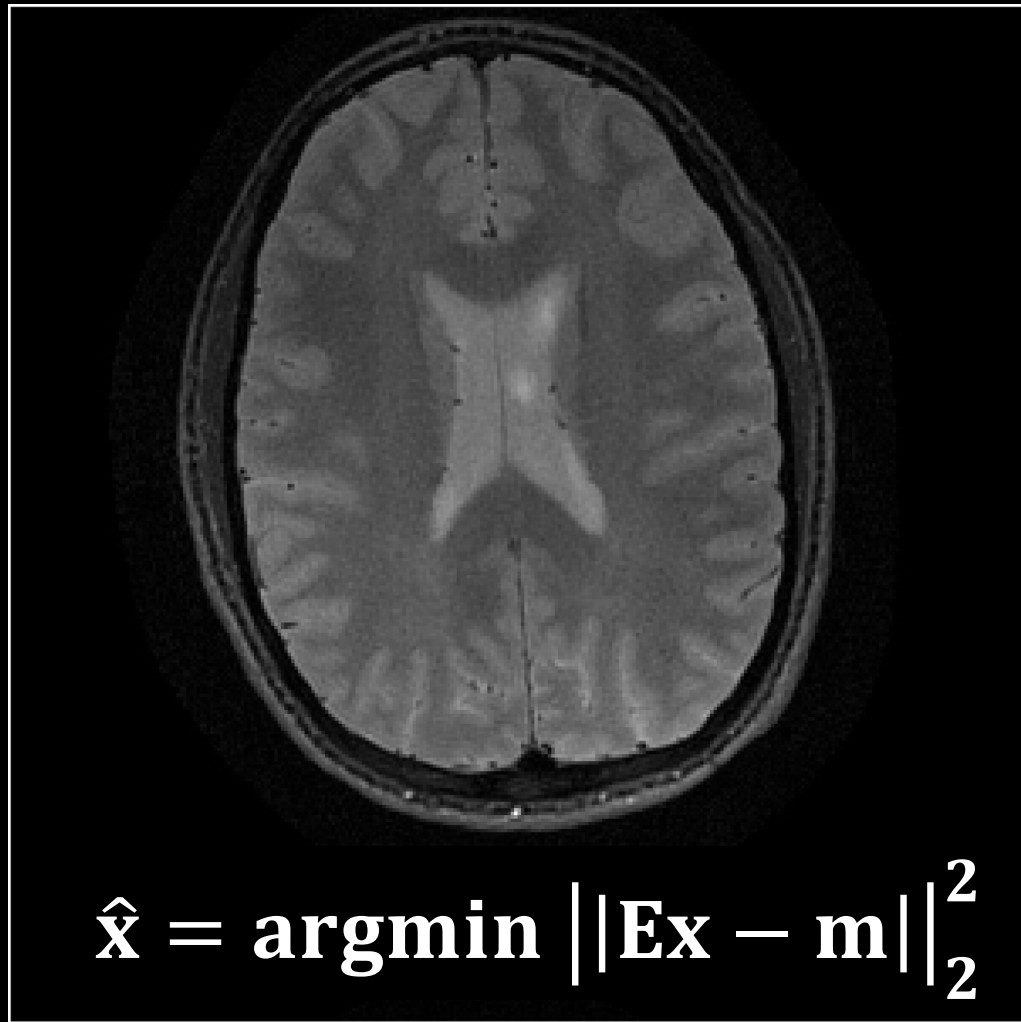


SENSE and GRAPPA are
for Cartesian k-space

while

CG-SENSE is for arbitrary
k-space trajectories

CG-SENSE reduces
streaking artifacts and
improves image sharpness
for radial MRI

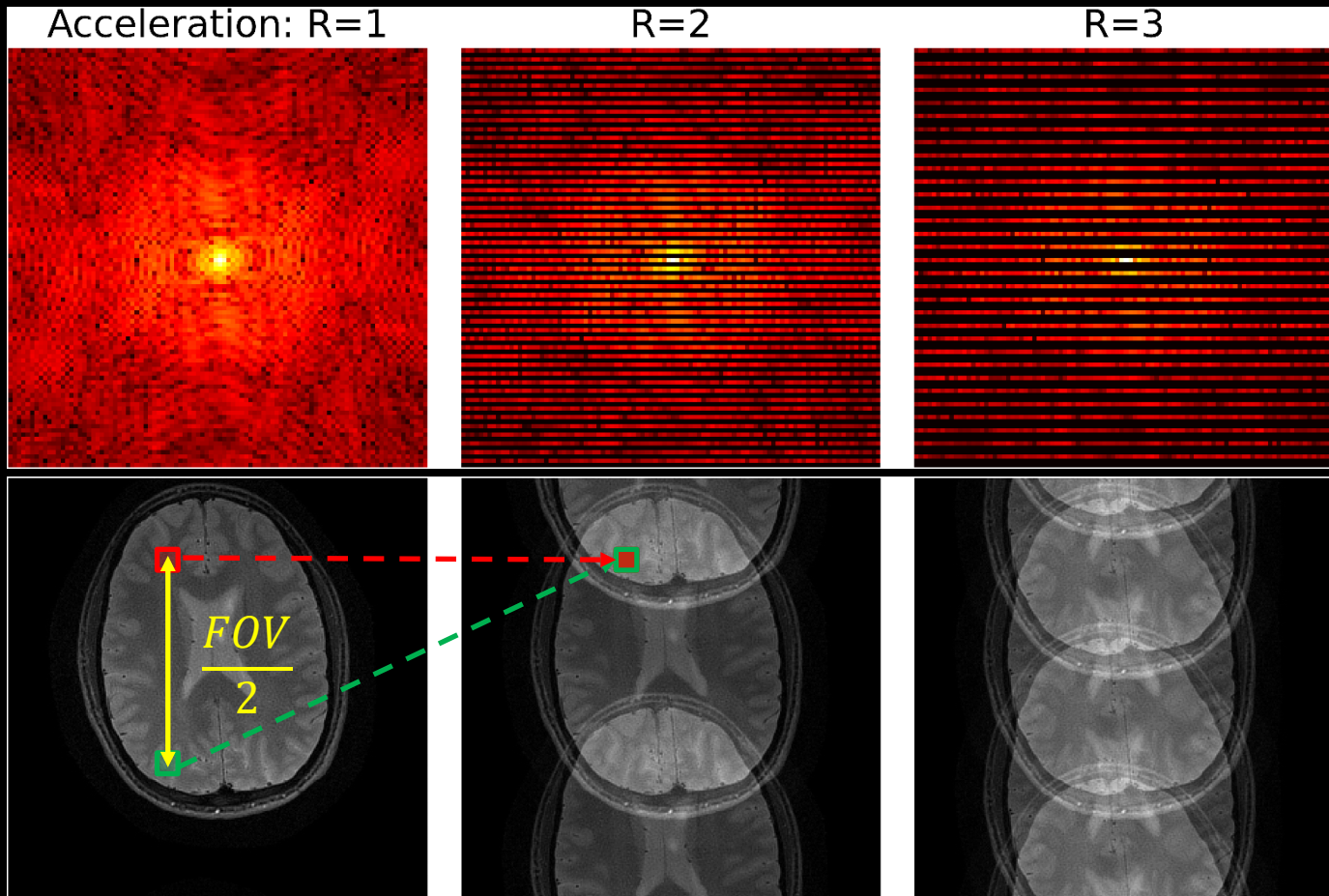


SENSE

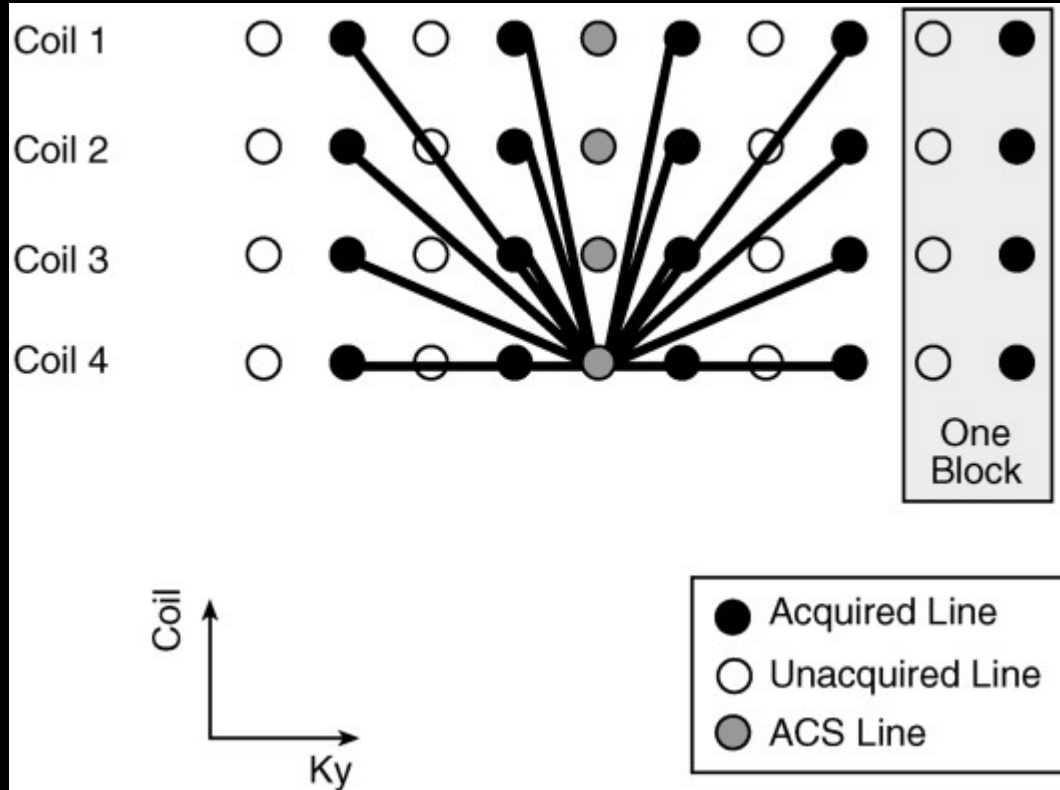
Sensitivity Encoding:
Parallel Imaging for
Cartesian k-space

Cartesian
k-space

FFT
reconstruction

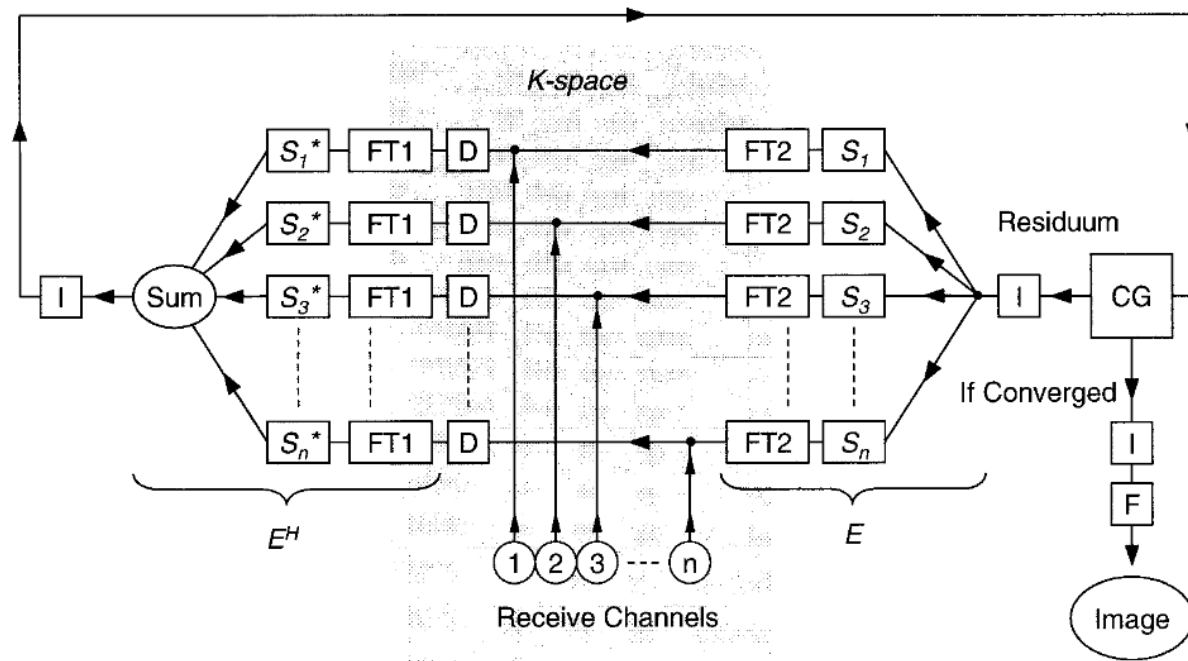


Generalized Autocalibrating Partially Parallel Acquisitions (GRAPPA) for Cartesian k-space data



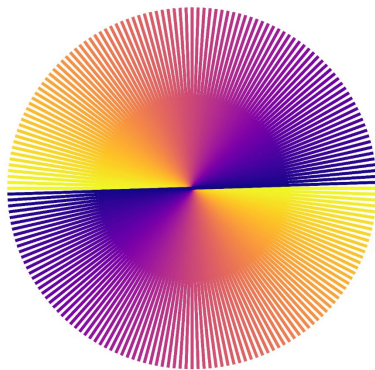
Conjugate Gradient SENSE reconstruction (CG-SENSE) for arbitrary k-space trajectories

SENSE With Arbitrary k-Space Trajectories

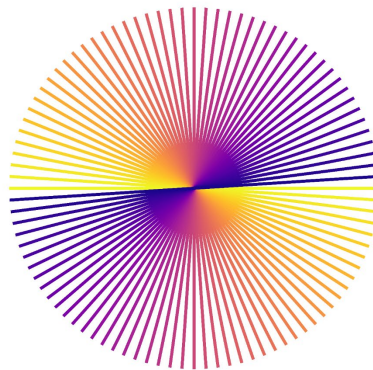


Radial
k-space
trajectory

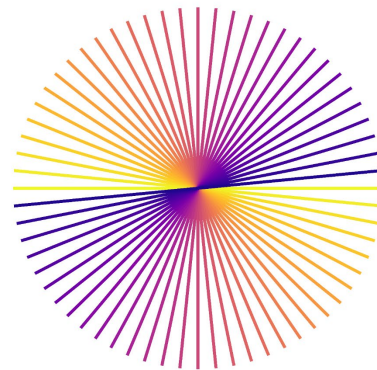
Acceleration: R=1
(96 spokes)



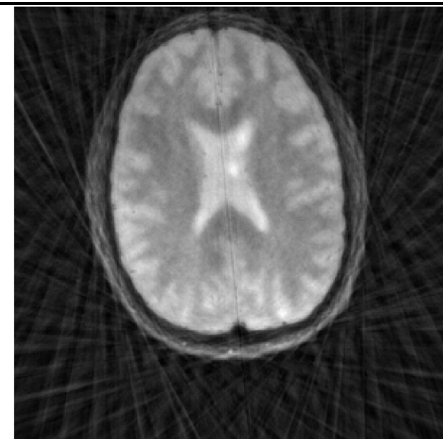
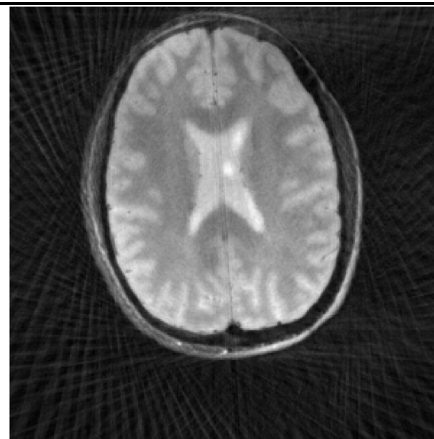
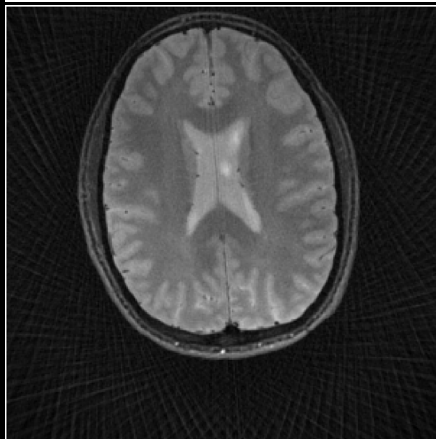
R=2
(48 spokes)



R=3
(32 spokes)

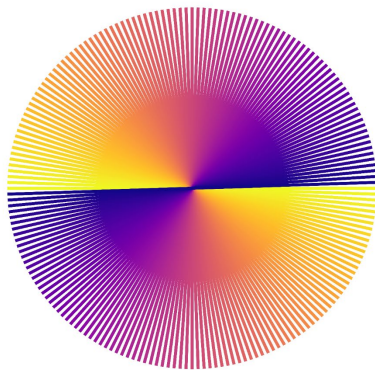


Gridding
reconstruction
(streaking &
blurring)

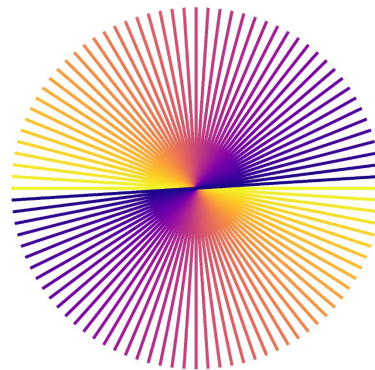


Radial
k-space
trajectory

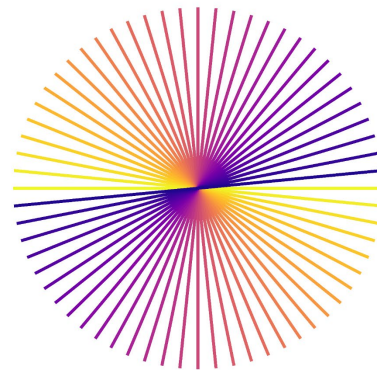
Acceleration: R=1
(96 spokes)



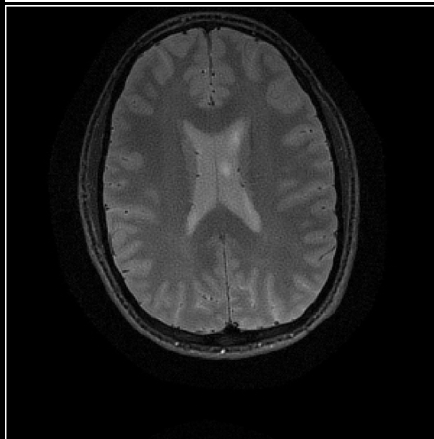
R=2
(48 spokes)



R=3
(32 spokes)



CG-SENSE
reconstruction



**Utilizing
CG-SENSE
in clinical
setting**

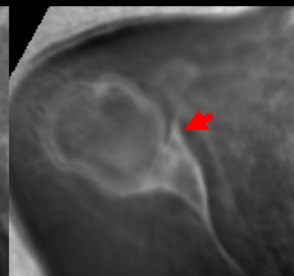
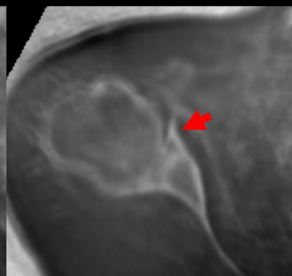
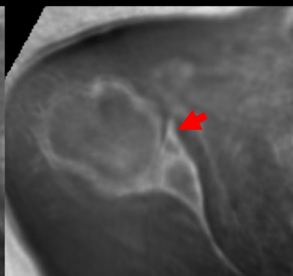
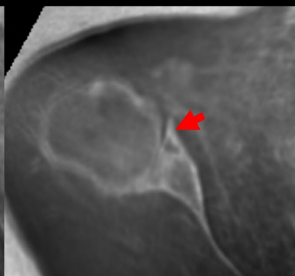
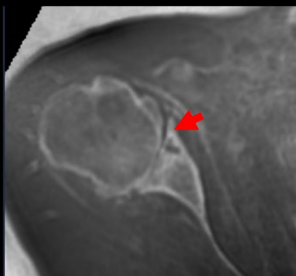
GRID: 5.8min

2.9min

1.9min

1.4min

1.2min



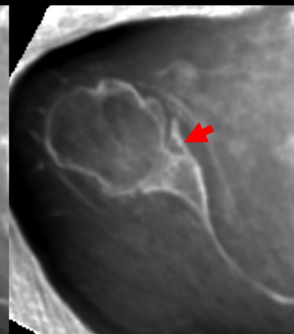
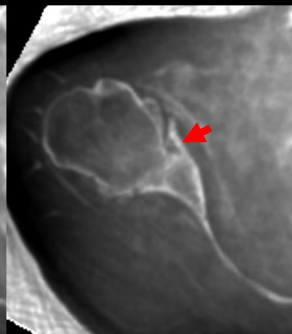
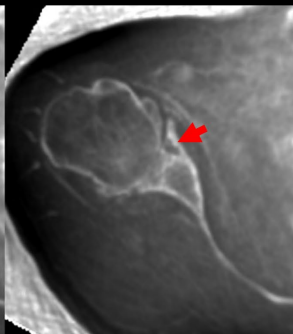
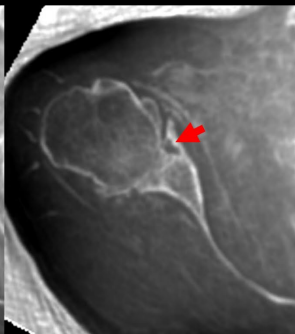
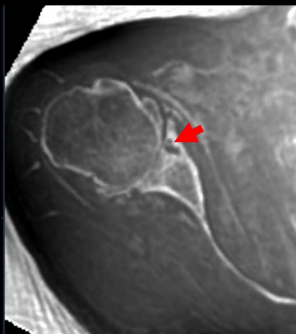
DLR: 5.8min

2.9min

1.9min

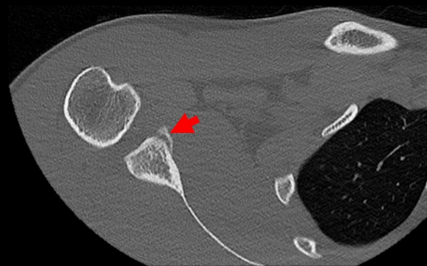
1.4min

1.2min



**Bony Bankart
Lesion
(arrow)**

CT



**Hung Do et al., ISMRM 2025,
Abstract #0156**

4-echo UTE, 0.8mm³
with CG-SENSE & Deep
Learning Denoising Recon
(DLR) vs. Gridding



ISMRM 2025 – Honolulu, HI

Want to learn more?

The Canon logo is displayed in its signature red color.

Ultra short TE multi-echo (UTE multi-echo)



Hung P. Do, PhD

WHAT?

An MRI pulse sequence that allows acquisition of multi-echo data with ultra short echo times (TEs).

WHY?

Ultra short TEs allows reception of signals from tissues with short $T2^*$ which can not be obtained with conventional field echo (gradient echo) sequences.

WHEN?

UTE multi-echo is available at both 1.5T and 3T and can be used to visualize tissues with short $T2^*$ and to generate associated $T2^*$ maps.

<https://global.medical.canon/products/magnetic-resonance/good-to-know#ultrashort>



Want to learn more?



Advanced intelligent Clear-IQ Engine (AiCE)



Hung P. Do, PhD

- WHAT?** A deep learning based reconstruction method for MRI that intelligently removes noise while maintaining feature integrity.
- WHY?** To increase SNR of the reconstructed images. This increased SNR could be translated to increased resolution and/or shortened scan time. This could also enable high field-like image quality without high-field challenges (e.g. higher cost, B₀ & B₁ inhomogeneity, etc.).
- WHEN?** Applicable to all anatomies and available at both 1.5T and 3T, for both wide and narrow bore system.

<https://global.medical.canon/products/magnetic-resonance/good-to-know#advanced>



Want to learn more?

The Canon logo is displayed in its characteristic red color and bold, sans-serif font.

Conjugate Gradient Reconstruction CG Recon



Chia-Ying Liu, PhD

- WHAT?** CG Recon is a reconstruction technique that allows parallel imaging to be applied for non-Cartesian sampling acquisitions
- WHY?** The aliasing that results from undersampled data is considerably more complex in non-Cartesian acquisitions and prevents straightforward unfolding by conventional parallel imaging methods
- WHEN?** CG Recon enables parallel imaging for radial ultra-short echo time (UTE) imaging

<https://global.medical.canon/products/magnetic-resonance/good-to-know#cgrecon>



Bonus: Challenging Artifacts

The Semi-convergence Behavior
of CG-SENSE

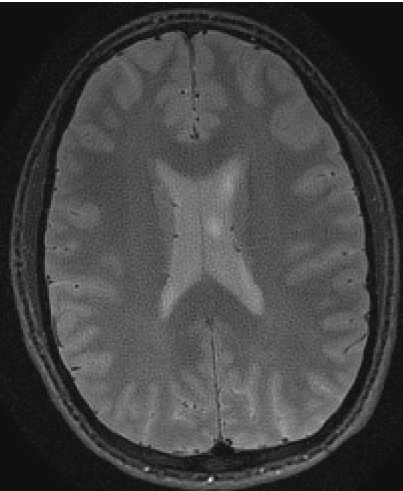


Noise amplification with a high number of iterations

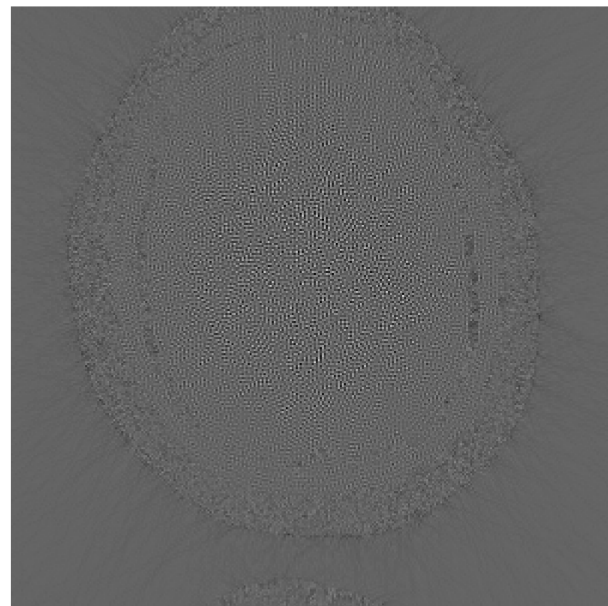
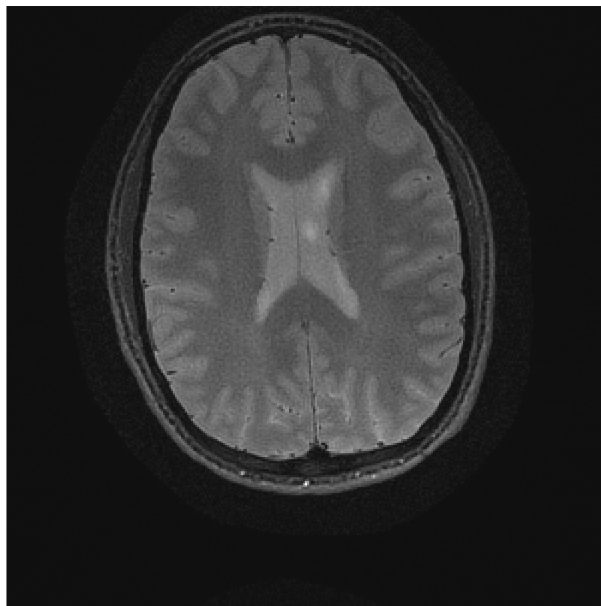
50th iteration

7th iteration

difference (10x)



Noise amplification



CG-SENSE's Semi-convergence

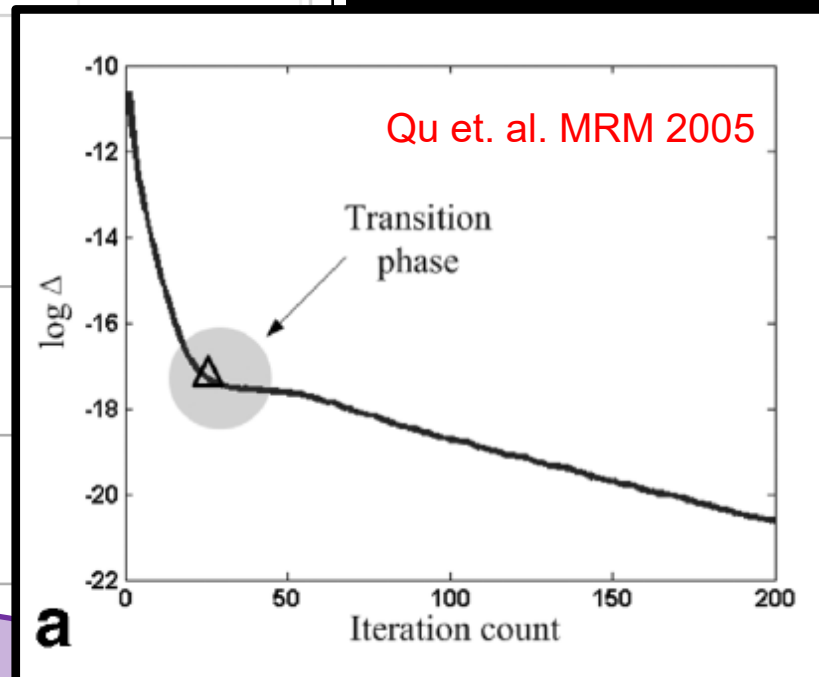
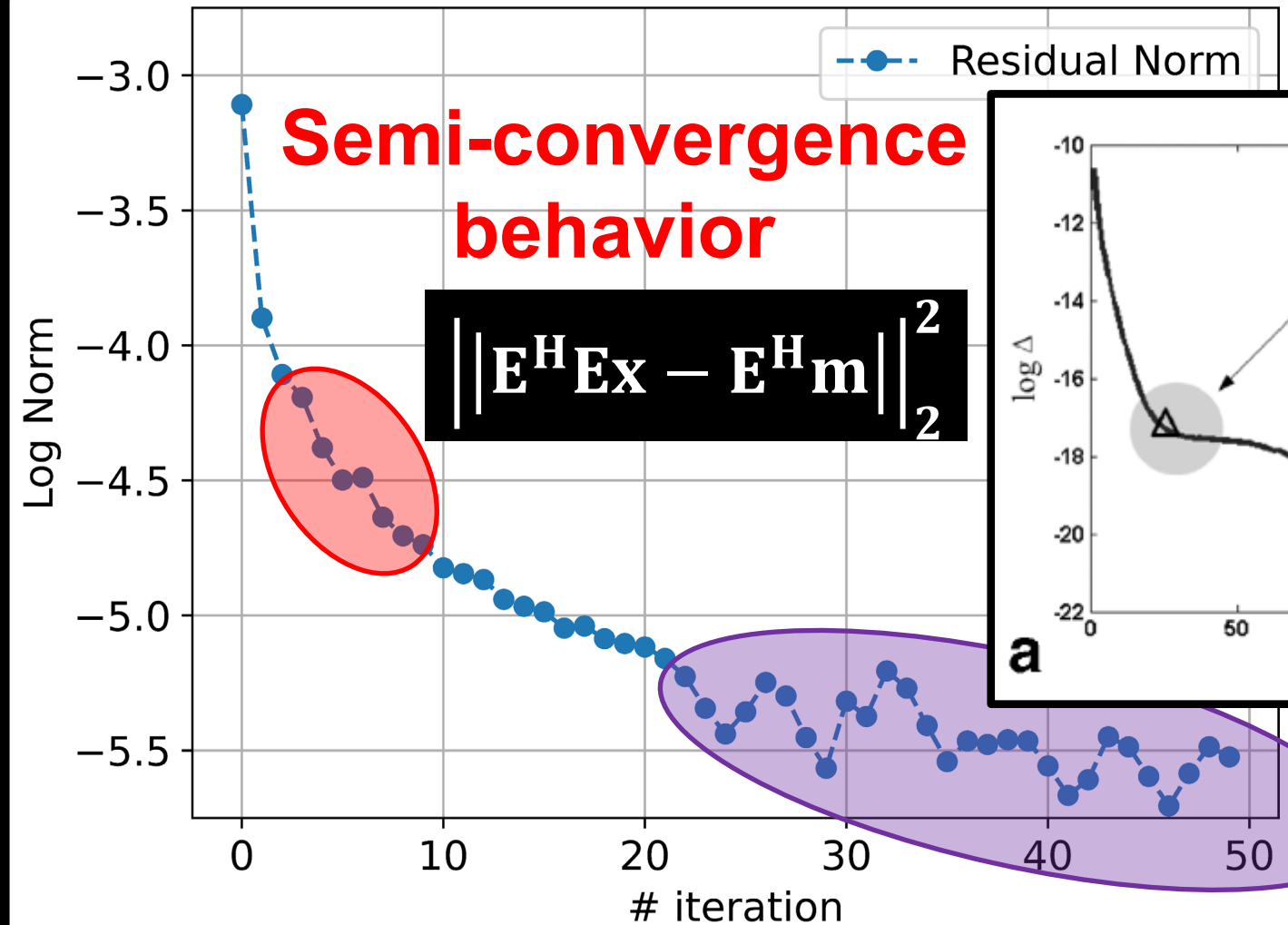
- “*Semi-convergence is characterized by initial convergence toward the optimal solution but later divergence*” (Qu et al., MRM 2005).
- The under-sampled k-space data causes the inverse problem ill-posed, which in turns leads to the semi-convergence behavior.
- The semi-convergence is a feature of the Conjugate Gradient or Gradient Descent algorithm when a problem is ill-posed.



Troubleshooting

1. My first response, when I saw the grainy (noise-like) artifacts, was to increase the number of iterations, hoping that the reconstructed image would come closer toward the optimal solution (un-aware of the *semi-convergence* behavior). However, the noise amplification got worse.
2. Second, I wondered if there is a bug in the CG implementation. I decided to reimplement different variants of the CG algorithm, but the artifacts remained.
3. Third, I implemented gradient descent algorithm, but the artifacts persisted.
4. Fourth, I inspected the intermediate images and plotted residual norm vs. number of iterations. Residual norm behaved funny at high iteration count, but I wasn't sure why.
5. Finally, I found the Qu *et al.*, MRM2005 and learned about the *semi-convergence* phenomenon.
6. The solution is to use early stopping or to implement CG with regularizations.

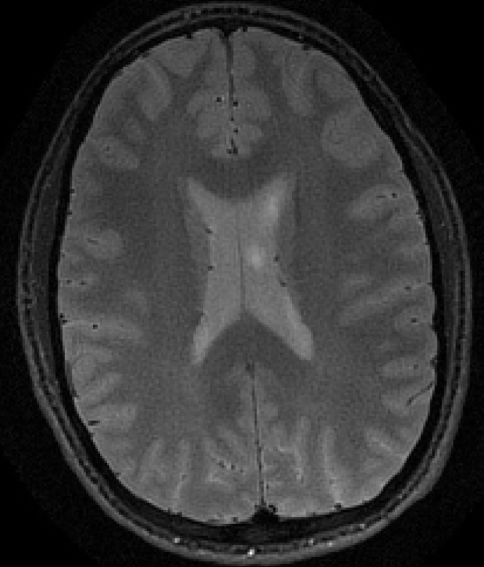
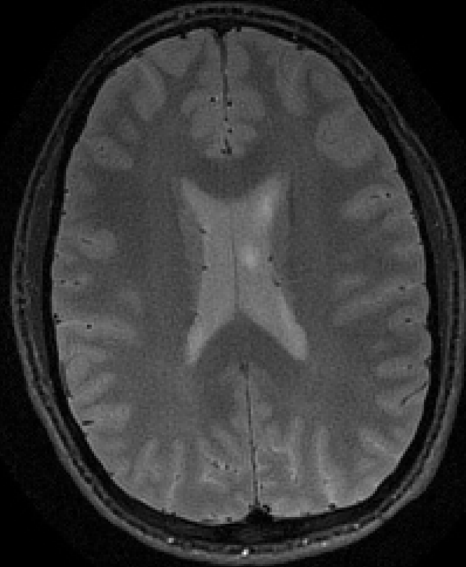
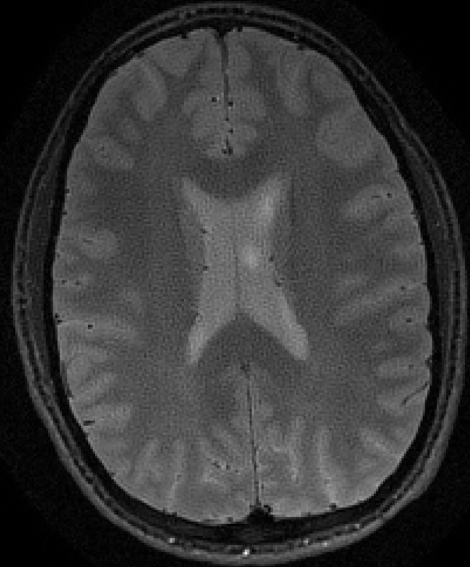




Noise
amplification

Tikhonov
regularization

Early stopping



$$\hat{\mathbf{x}} = \operatorname{argmin} \left\| \mathbf{E}\mathbf{x} - \mathbf{m} \right\|_2^2 + \frac{\lambda}{2} \left\| \mathbf{x} \right\|_2^2$$



Other Regularizations

Early stopping and Tikhonov regularization mitigate the noise amplification; other advanced regularizations also work:

- L1-regularization in the sparsifying transform domain as in Compressed Sensing reconstruction,
- Data-driven regularization as in Deep Neural Network-based reconstruction, etc.

$$\hat{\mathbf{x}} = \mathbf{argmin} \left\| \mathbf{E}\mathbf{x} - \mathbf{m} \right\|_2^2 + \mathbf{R}(\mathbf{x})$$

$\mathbf{m}$: measured k-space

$\mathbf{E}$: encoding matrix

$\mathbf{x}$: intermediate reconstructed image

$\mathbf{R}(\mathbf{x})$: regularization function



References

1. Qu, Peng, et al. "Convergence behavior of iterative SENSE reconstruction with non-Cartesian trajectories." *Magnetic Resonance in Medicine* 54.4 (2005): 1040-1045.
2. Pruessmann, Klaas P., et al. "Advances in sensitivity encoding with arbitrary k-space trajectories." *Magnetic Resonance in Medicine* 46.4 (2001): 638-651.
3. Maier, Oliver, et al. "CG-SENSE revisited: Results from the first ISMRM reproducibility challenge." *Magnetic Resonance in Medicine* 85.4 (2021): 1821-1839.
4. Griswold, Mark A., et al. "Generalized autocalibrating partially parallel acquisitions (GRAPPA)." *Magnetic Resonance in Medicine* 47.6 (2002): 1202-1210.
5. Lustig, Michael, David Donoho, and John M. Pauly. "Sparse MRI: The application of compressed sensing for rapid MR imaging." *Magnetic Resonance in Medicine* 58.6 (2007): 1182-1195.
6. Hammernik, Kerstin, et al. "Learning a variational network for reconstruction of accelerated MRI data." *Magnetic resonance in medicine* 79.6 (2018): 3055-3071.
7. Professor John Pauly's Lecture on "Reconstruction of Non-Cartesian Data", Stanford University
8. Pruessmann, Klaas P., et al. "SENSE: sensitivity encoding for fast MRI." *Magnetic Resonance in Medicine: An Official Journal of the International Society for Magnetic Resonance in Medicine* 42.5 (1999): 952-962.





ISMRM & ISMRT
ANNUAL MEETING & EXHIBITION

Honolulu, Hawai'i, USA 10-15 MAY 2025



ISMRM 2025

MR Artifacts Game Show

**Classic Artifacts in Radial MRI and
The Semi-convergence Behavior of CG-SENSE**

Hung Do, PhD MSEE
Canon Medical Systems USA



ISMRM 2025 – Honolulu, HI